Optimizing neural network architectures using Genetic Algorithms, Particle Swarm Optimization, and Simulated Annealing

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Optimization Methods for Engineers, August 2025

Overview

Introduction - Problem Description

- **Given** an image-based dataset with n data points $d_1, d_2, ..., d_n \in \mathbb{R}^{H \times W \times C}$, where H, W, C represent height, width and channels respectively, and $n \in \mathbb{N}$ represents the number of samples.
- **Find** $\alpha = (\alpha_1, \alpha_2, ..., \alpha_n)^\top \in \mathcal{D}^n$ which maximizes

$$f(\alpha, \mathcal{D}) := \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha)$$

Where:

- α represents the network architecture parameters
- $\mathcal{D}$ is the dataset
- λ_1, λ_2 are weighting factors balancing size vs. accuracy
- NetworkSize(α) counts the total number of parameters
- Accuracy($\alpha, \mathcal{D}$) is the classification accuracy on dataset $\mathcal{D}$

Introduction - Motivation

- For L maximum layers and C component types, exhaustive search requires $\mathcal{O}(C^L)$ evaluations
- **Consequence:** Analytical methods become computationally unviable
- **Solution:** Metaheuristic algorithms (GA, PSO, SA) provide more efficient exploration [1–5]
- **Expansion:** Consider more layers such as Conv1D/Conv2D, that were not considered in the related work

Introduction - Architecture Components

Unlike traditional approaches that only focus on fully connected layers, we extend the search space to include the following layers:

- **Convolutional Layers (Conv1D/Conv2D):** Feature extraction with learnable filters
- **Fully Connected Layers (FC):** Dense layers for classification
- **Batch Normalization:** Normalization of the values for training stability
- **Max Pooling:** Spatial dimension reduction
- **Dropout:** Regularization technique to avoid overfitting
- **Activation Functions:** ReLU, Tanh, Sigmoid

Introduction - Solution Space Characteristics

- Non-linear cost function
  $f(\alpha, \mathcal{D}) := \lambda_2 \cdot \text{Accuracy}(\alpha, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha)$ combines discrete network size with continuous accuracy metrics
- Mixed discrete-continuous solution space consisting of architecture topology (discrete) and hyperparameters (continuous)
- Solution space can be represented by encoding each architecture α as a variable-length chromosome
- **Naive Approach:** Exhaustive search with runtime $\mathcal{O}(C^L)$ for L max layers with C component types becomes intractable for realtistic network sizes

Introduction - Project Objectives

- Investigate how **Genetic Algorithms (GA)**, **Particle Swarm Optimization (PSO)**, and **Simulated Annealing (SA)** perform when applied to the problem.
- Balance the trade-off between network size and accuracy
- Evaluate the effectiveness of different layer types in automated architecture design
- Assess the performance of these algorithms by comparing them to baseline methods, such as **Random Search (RS)** and **Grid Search (GS)**.
- Assess the performance of the algorithms across different dataset complexities

Algorithms - Baseline Methods

Given a image-based dataset $\mathcal{D} = \{d_1, ..., d_n\}$

Random Search (RS)
Cost function: $f(\alpha^{(i)}) = \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)})$

1. Generate N random candidate architectures $\alpha^{(1)}, ..., \alpha^{(N)}$
2. Train each network and evaluate $f(\alpha^{(i)})$ for $i = 1, 2, ..., N$
3. Return $(\alpha^{(best,RS)}, f(\alpha^{(best,RS)}))$ with $f(\alpha^{(best,RS)}) \geq f(\alpha^{(i)}), \forall \alpha^{(i)}$

Grid Search (GS)
Cost function: $f(\alpha^{(i)}) = \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)})$

1. Define systematic grid over architecture hyperparameters
2. Evaluate $f(\alpha^{(i)})$ for each grid point $\alpha^{(i)}$
3. Return $(\alpha^{(best,GS)}, f(\alpha^{(best,GS)}))$ with maximal fitness

Algorithms - Genetic Algorithm

Genetic Algorithm (GA) [1]:

Fitness function: $\textit{fitness} = \lambda_2 \cdot \text{Accuracy}(\alpha^{(i)}, \mathcal{D}) - \lambda_1 \cdot \text{NetworkSize}(\alpha^{(i)})$

Encoding: Variable-length chromosomes representing layer sequences

- Gene type: {Conv2D, Conv1D, FC, BatchNorm, MaxPool, Dropout, Activation}
- Gene parameters: kernel_size, stride, filters, units, etc.

Algorithm Steps:

1. Create `population_size` random architectures $\alpha^{(i)} = (\textit{layer}_1, \textit{layer}_2, ..., \textit{layer}_k)$
2. Train networks and compute fitness scores
3. Tournament selection with `tournament_size` individuals
4. Single-point crossover with probability `crossover_prob`, preserving valid layer sequences
5. Add/remove layers with probability `mutation_add_prob` / `mutation_del_prob` and modify layer parameters with probability `mutation_param_prob`
6. Generate new population, repeat for `num_generations` generations
7. Return $(\alpha^{(\textit{best},GA)}, \textit{fitness}^{(\textit{best},GA)})$

Algorithms - Particle Swarm Optimization

Particle Swarm Optimization (PSO) [3, 4]:

Position Representation: Continuous vector $x^{(i)} \in \mathbb{R}^d$ encoding the architecture parameters

Velocity Update:

$$v^{(i)}(t+1) = w \cdot v^{(i)}(t) + c_1 \cdot r_1 \cdot \left(pbest^{(i)} - x^{(i)}(t)\right) + c_2 \cdot r_2 \cdot \left(gbest - x^{(i)}(t)\right)$$

Position Update: $x^{(i)}(t+1) = x^{(i)}(t) + v^{(i)}(t+1)$

Architecture Decoding: Map continuous positions to discrete architectures:

- Layer count: $\lfloor x_1 \cdot \texttt{max\_layers} \rfloor$
- Layer types: $\text{argmax}\left(\text{softmax}\left([x_2, x_3, ..., x_7]\right)\right)$
- Layer parameters: sigmoid/tanh transformations to valid ranges

Algorithm Steps:

1. Initialize swarm: Set positions $x^{(i)}(0)$ at random and set velocities $v^{(i)}(0) = 0$
2. Convert positions to architectures and train the networks
3. Update personal best $pbest^{(i)}$ and global best $gbest$
4. Update velocities and positions
5. Repeat for `num_iterations` iterations
6. Return $(\alpha^{(best,PSO)}, fitness^{(best,PSO)})$

Algorithms - Simulated Annealing

Simulated Annealing (SA) [2]:

State Representation: Current architecture $\alpha_{current}$
Neighbourhood Function: Small architectural modifications:

- Add/remove single layer
- Modify layer parameters
- Swap adjacent layers

Algorithm Steps:

1. Start with random architecture α_{current}
2. Apply random modification to generate $\alpha_{\text{neighbour}}$
3. Acceptance Criterion:
   - If $f(\alpha_{\text{neighbour}}) < f(\alpha_{\text{current}})$: accept
   - Else: Accept with probability $e^{-\left(\frac{f(\alpha_{\text{neighbour}}) - f(\alpha_{\text{current}})}{T}\right)}$
4. Reduce temperature $T = T \cdot \texttt{cooling\_rate}$ every 10 iterations
5. Repeat until convergence or `max_iterations`
6. Return $(\alpha^{(best,SA)}, fitness^{(best,SA)})$

Parametric Optimization - Variable Parameters

All algorithms run on variable parameters:

- Variable parameters for GA:

`pop_size`	`cross_prob`	`mut_add`	`mut_del`	`mut_param`	`tourn_size`	`n_gen`
Population size	Crossover prob.	Add mutation	Del mutation	Param mutation	Tournament size	Generations

- Variable parameters for PSO:

`swarm_size`	`w`	c_1	c_2	`num_iter`
Swarm size	Inertia weight	Cognitive coefficient	Social coefficient	Iterations

- Variable parameters for SA:

`initial_temp`	`cooling_rate`	`max_iter`
Initial temperature	Cooling rate	Max. iterations

- Other degrees of freedom include: the type of optimizer used in the neural network, learning rate used in training, type of crossover used in the GA, choice of r_1 and r_2, etc.

Parametric Optimization - Fixed Parameters

- General fixed parameters used in this work:
  - $\lambda_1 = 10^{-6}$ (parameter penalty)
  - $\lambda_2 = 1.0$ (accuracy weight)
  - `max_layers` $= 8$
  - `epochs` $= 3$ (training) and 20 (evaluation)
  - `learning_rate` $= 0.001$
  - `optimizer` $=$ Adam with `weight_decay` $= 10^{-4}$
- Algorithm-specific fixed parameters used in this work:
  - GA: `pop_size` $= 40$, `n_gen` $= 30$
  - PSO: `swarm_size` $= 40$, `num_iter` $= 30$
  - SA: `max_iter` $= 1200$

Parametric Optimization - Optimization Methodology

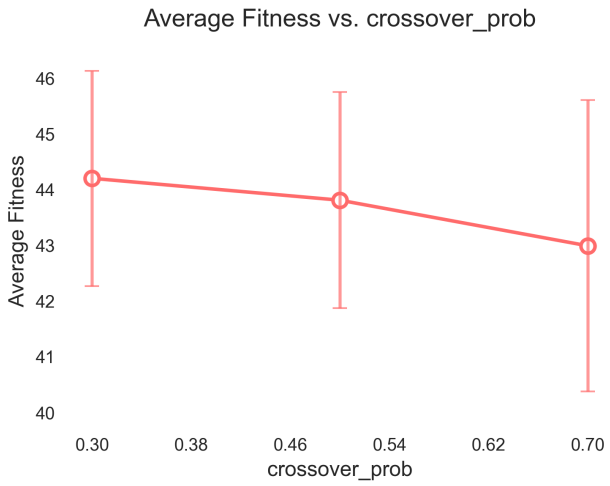
Simple algorithm to find the best parameters:

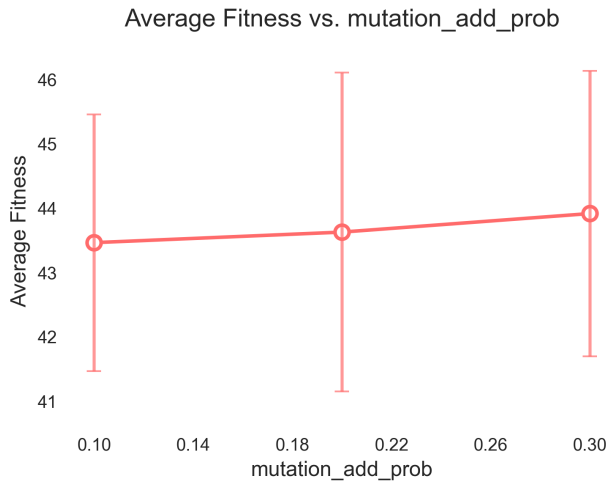
for each algorithm $\in$ {GA, PSO, SA} **do**
  for each parameter_set $\in$ parameter_combinations **do**
    for $k = 1$ to 3 **do**
      Split dataset into train/validation
      Run algorithm with parameter_set
      Collect: best_fitness, convergence_time, final_accuracy
    end for
    Compute: avg_fitness, avg_time, avg_accuracy
  end for
  Select parameter_set with best avg_fitness
end for

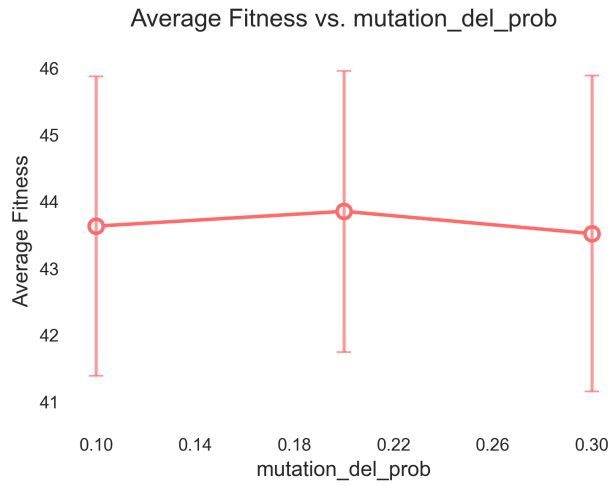
Dataset (CIFAR10 / MNIST) Splits:

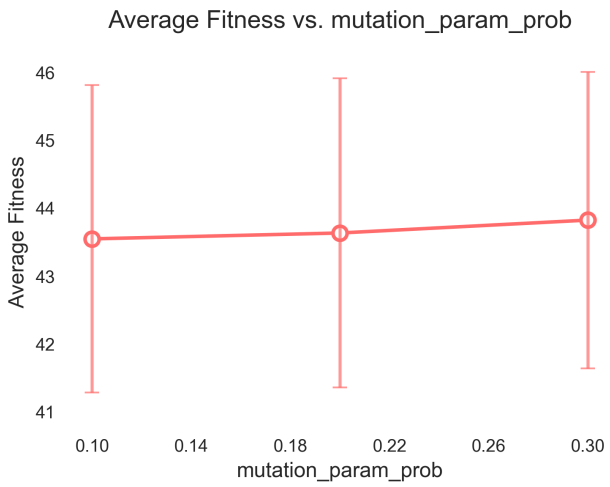
- Training: 60% (Architecture search)
- Validation: 20% (Fitness evaluation)
- Test: 20% (Final performance assessment)

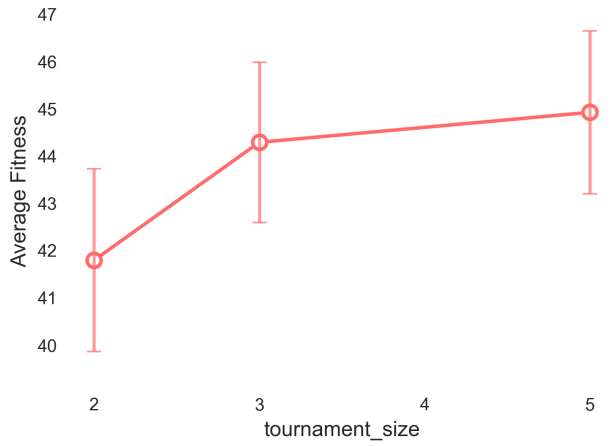
Parametric Optimization - Genetic Algorithm





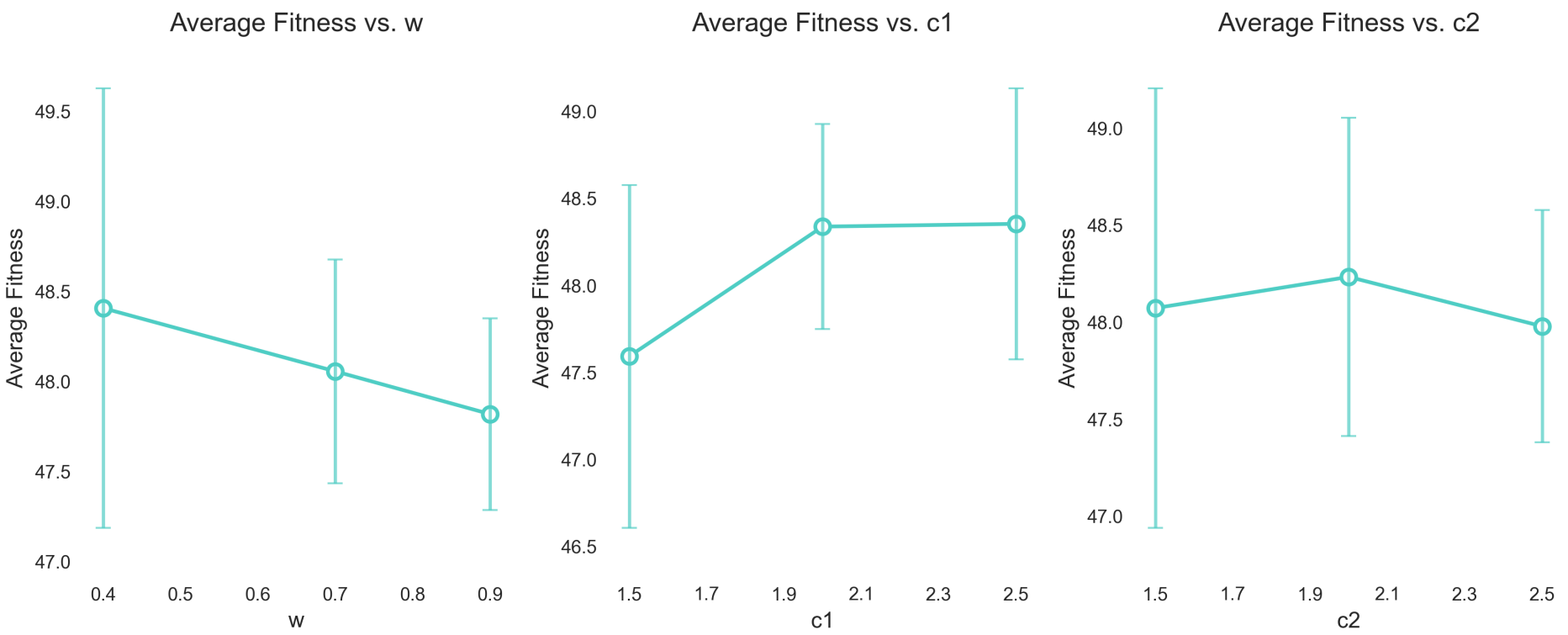






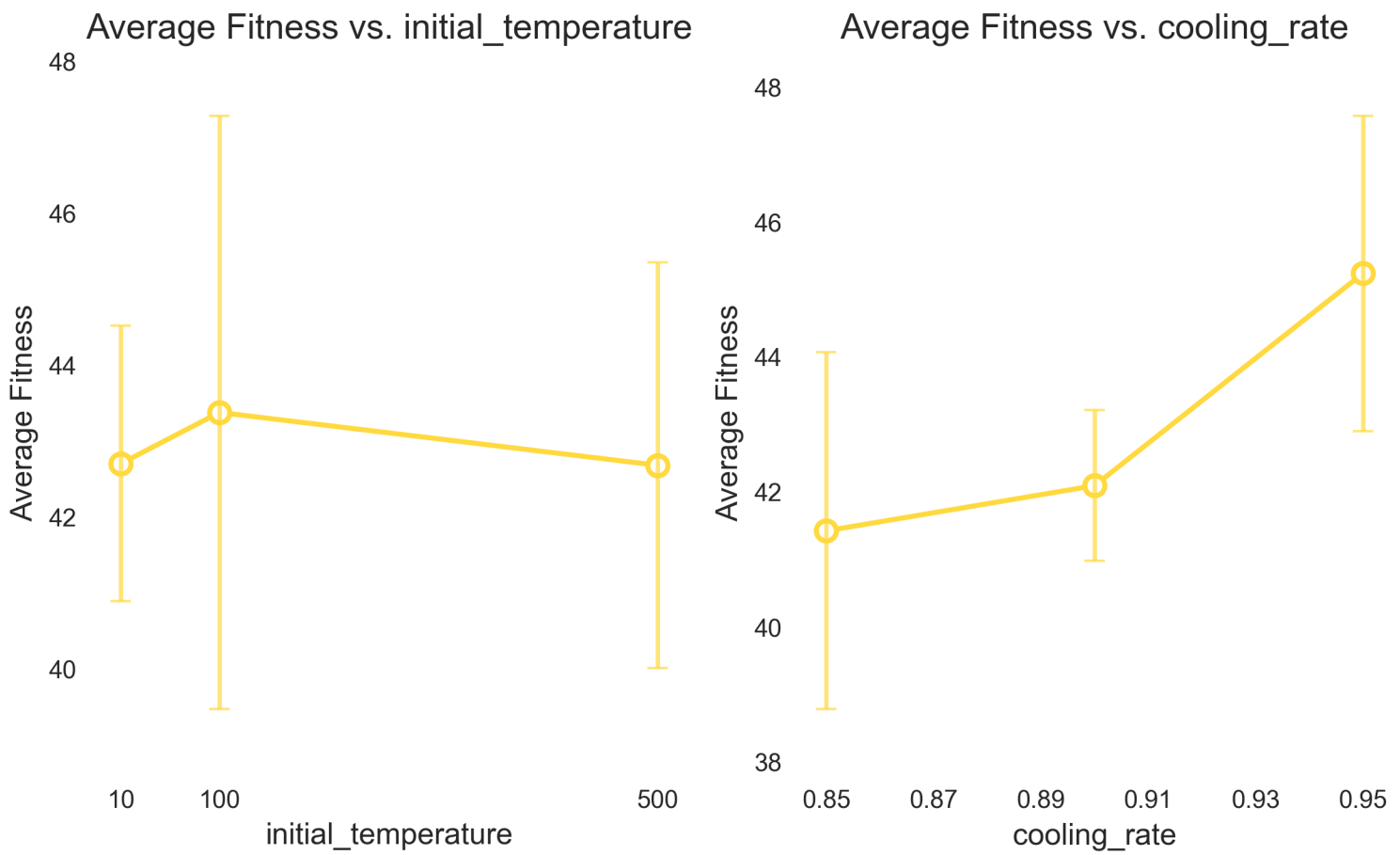
- In conclusion, the following values were chosen: $\texttt{cross\_prob} = 0.3$, $\texttt{mut\_add} = 0.3$, $\texttt{mut\_del} = 0.2$, $\texttt{mut\_param} = 0.2$, $\texttt{tourn\_size} = 5$

Parametric Optimization - Particle Swarm Optimization



- In conclusion, the following values were chosen: $\texttt{w} = 0.4$, $\texttt{c1} = 2.5$, $\texttt{c2} = 2.0$

Parametric Optimization - Simulated Annealing



- In conclusion, the following values were chosen: `initial_temp` = 100, `cooling_rate` = 0.95

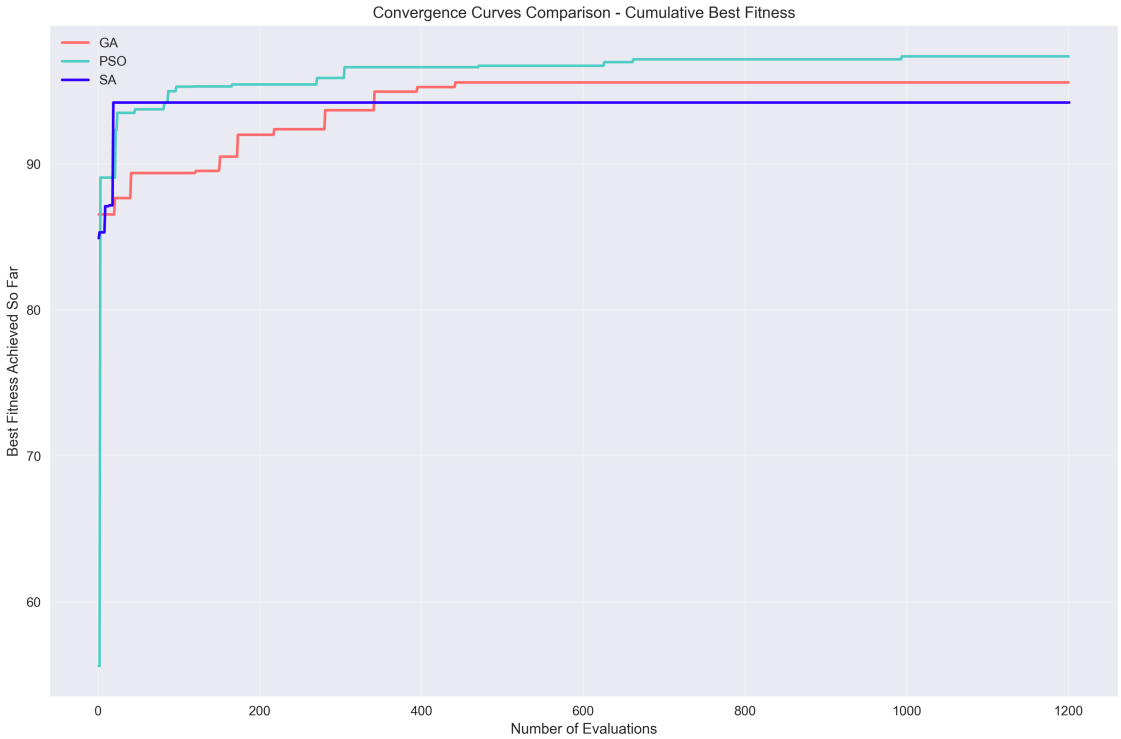
Results - Overview

Algorithm	Test Accuracy (%)	Parameters	Fitness	Runtime (min)
GA	97.2%	213k	95.412	7.2
PSO	97.5%	146k	97.354	14.3
SA	96.0%	27k	94.191	18.2

Table: Algorithm Performance Comparison on MNIST

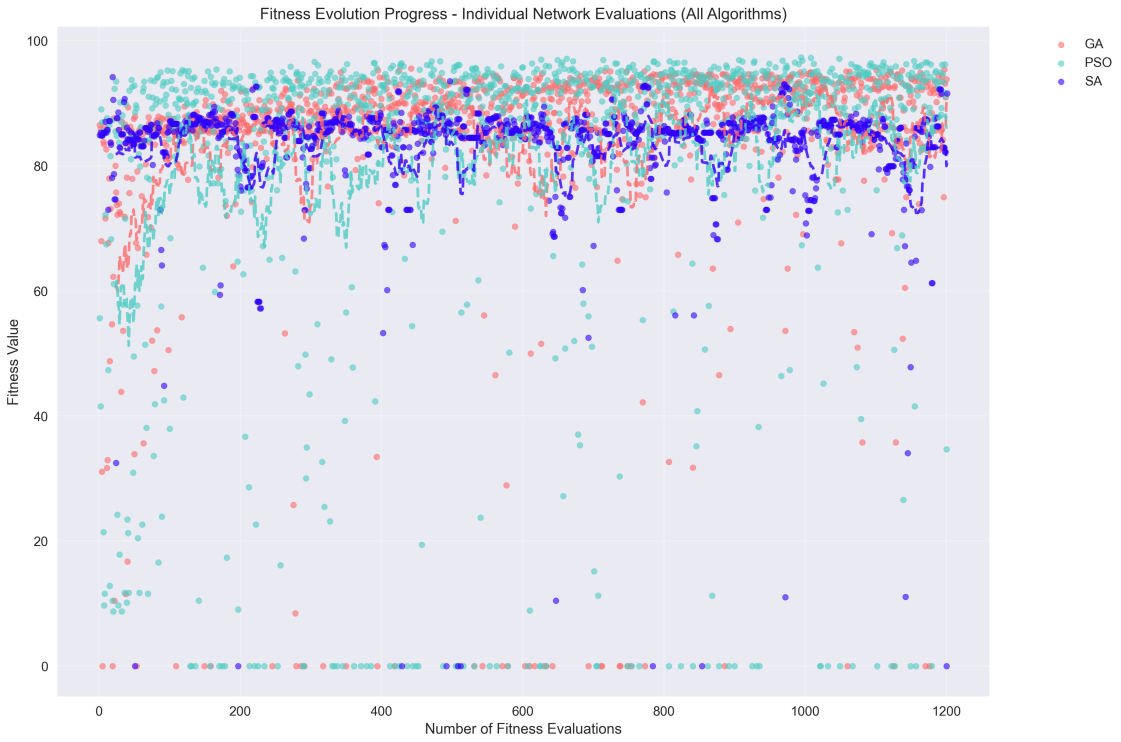
- All of the tested algorithms show promising results, with a test accuracy of $\geq 96\%$
- The most accurate model was produced by PSO
- However the most accurate model is not the largest model (146k vs. 213k parameters)
- The runtime of both GA and PSO is lower than SA, as they both could be easily implemented with parallelization in mind

Results - Convergence



- GA shows rapid initial convergence but plateaus early
- PSO demonstrates steady improvement throughout iterations
- SA exhibits characteristic cooling behavior with gradual convergence
- All algorithms converge within computational budget

Results - Fitness Evaluation



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Conclusion and Discussion

- **Genetic Algorithm (GA):**
  - Crossover seems to have a negative influence
  - More chance to add layers has a positive effect.
  - $\Longrightarrow$ To be expected, since larger networks should perform better
- **Particle Swarm Optimization (PSO):**
  - PSO consistently performed the best for this task across most experiments and is not very volatile to parameter changes.
- **Simulated Annealing (SA):**
  - SA performed the worst and can also not be parallelized like the other algorithms.
  - We would not recommend using SA to optimize neural network architectures.
- Experiments were overall success and improvement over simpler methods such as Random Search and Grid Search

Future Work and Limitations

Current Limitations:

- Limited to relatively simple architectures ($\leq$ 10 layers)
- Training epochs are reduced for computational feasibility
- Single-objective scalarization of multi-objective problem
- Runtime of certain approaches became infeasible very quickly

Future Extensions:

- Generalize the approach to different datasets (graph, audio, video, etc)
- Introduce more layers to be considered when constructing the NNs
- Consider advanced components, such as attention, transformers, etc.
- Leverage pre-trained components in search

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